



YPDC

Integrated Portal System

Complete System Requirements, Functional Design
& Implementation Report

STUDENT • MEMBER • DIRECTORATE • FACULTY • ADMIN • ALUMNI • PUBLIC
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7 PORTALS	81 MODULES	6 USER ROLES	v1.0 VERSION
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Document Title	YPDC Integrated Portal System — System Report
Version	1.0 • Initial Release
Date	10 September 2026
Classification	Official — For YPDC Leadership & Development Team
Coverage	Student • Member • Directorate • Faculty • Admin • Alumni • Public
Status	Approved for Planning & Development

This report defines the complete functional scope, module specifications, roles & permissions, workflows, reporting framework and implementation roadmap for the YPDC digital platform.

Inside: Executive Summary • 7 Portal Specifications (81 modules) • Roles & Workflows • Data Model • Analytics • Roadmap • Checklists & Glossary

PREPARED FOR YPDC LEADERSHIP • SEPTEMBER 2026

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1. Executive Summary

The **YPDC Integrated Portal System** unifies seven specialised portals — **Student, Member, Directorate, Faculty, Admin, Alumni and Guest/Public** — into one secure, role-based digital platform. Together they cover the complete lifecycle of engagement with YPDC: a visitor discovers the organisation on the public website, registers as a volunteer or member, participates in events, takes on directorate responsibilities, gets mentored by faculty, earns certificates and achievements, and finally continues as alumni — mentor, employer or supporter.

7
PORTALS

81
MODULES

15+
WORKFLOWS

100+
REPORT KPIS

This report specifies all **81 functional modules**, the **roles & permissions model**, **cross-portal workflows** (membership, events, tasks, approvals, attendance, certificates, grievances), the **reporting framework**, **non-functional requirements** and a **phased implementation roadmap**. It is written to serve three audiences at once: **leadership** (for decisions), **developers** (for building), and **auditors** (for verification).

Key decisions this report enables

- ▶ **Scope lock:** exactly what each portal contains — no ambiguity during development.
- ▶ **Role model:** who can see and do what — enforced by 200+ granular permissions.
- ▶ **Workflow standards:** one approved way for proposals, approvals, attendance and certificates.
- ▶ **Build order:** Phase 1 (foundation + public site + admin) → Phase 4 (alumni & analytics) — value delivered early.

Reading guide

Sections 2-3 give context and architecture. Sections 4-10 specify each portal module-by-module. Sections 11-14 define cross-cutting systems, data, workflows and analytics. Sections 15-18 cover quality, roadmap, risks and conclusions. Appendix A is the master 81-module checklist.

2. About YPDC & Document Purpose

About YPDC

YPDC (Youth Peace & Development Council) is a youth-led organisation dedicated to peace-building, leadership development, community service and skills empowerment. It operates through specialised **directorates**, guided by **faculty mentors**, powered by **student and member volunteers**, and sustained by an **alumni network** — all visible to the public through an open, transparent platform.

Vision & Mission (platform reflection)

- ▶ **Vision:** an empowered generation of youth leading peaceful, developed and inclusive communities.
- ▶ **Mission:** to organise, train and recognise youth through structured programs, transparent governance and digital-first operations.
- ▶ **Platform promise:** every contribution is recorded, every achievement is verifiable, every process is traceable.

Document purpose & scope

In scope: all 7 portals, 81 modules, roles, workflows, reports, notifications, documents and grievance handling. Out of scope (for v1.0): mobile native apps (responsive web first), payroll/finance ERP, and third-party LMS integration — all planned as Phase-5 extensions.

Stakeholders

Stakeholder	Interest & responsibility
YPDC Leadership / Patrons	Vision, approvals, strategic oversight
Directors & Assistant Directors	Directorate planning, tasking, event ownership
Members & Students	Participation, duties, volunteering, feedback
Faculty Mentors / Advisors	Guidance, supervision, approvals, evaluations
Alumni	Mentorship, careers, networking, support
Admin / IT Team	Platform operations, content, compliance, support
Public / Guests	Discovery, trust, registration, enquiries

3. System Overview & Architecture

The seven portals at a glance

#	Portal	Primary users	Modules	Core purpose
1	Student Portal	Enrolled students	10	Participate, volunteer, earn recognition
2	Member Portal	Inducted members	10	Duties, tasks, performance & reporting
3	Directorate Portal	Directors, ADs, teams	13	Plan, execute & govern directorate work
4	Faculty Portal	Faculty / mentors	9	Guide, supervise & approve
5	Admin Portal	Administrators	15	Operate, govern & analyse everything
6	Alumni Portal	Graduates / ex-members	10	Network, mentor, hire & support
7	Guest / Public Portal	Everyone (no login)	14	Showcase, inform & convert visitors

Access model

- ▶ **Single Sign-On (SSO):** one account, multiple roles — a member who is also a student sees both portals via a role switcher.
- ▶ **Public by default, private by role:** Guest portal needs no login; all other portals enforce authentication + role checks on every page and API.
- ▶ **Digital identity:** every member/student carries a QR-coded digital ID; every certificate carries a verification code checked on the public site.

High-level architecture (recommended)

- 1. Presentation layer:** responsive web app (mobile-first) + public website sharing one design system; portals as role-routed workspaces.
- 2. Application layer:** modular services — Identity & Access, Membership, Events, Tasks, Attendance, Certificates, Notifications, CMS, Reports, Grievance.
- 3. Data layer:** relational database (primary records) + object storage (documents/media) + search index (directory, FAQs, publications).
- 4. Integration layer:** email/SMS/push gateways, QR generation & scanning, calendar feeds, export (PDF/Excel), and audit logging on every write.

Recommended technology principles (stack-agnostic)

- ▶ Modern web framework with server-side rendering for the public site (SEO) and app-style routing for portals.
- ▶ REST/JSON APIs with token authentication; full audit trail; automated backups and role-based data isolation.
- ▶ Accessibility (WCAG 2.1 AA), Urdu + English bilingual UI, and print-friendly outputs for all official documents.

4. Portal 1 — Student Portal

For enrolled students — participation, learning & recognition • **10 modules**

The Student Portal is the primary gateway for enrolled students to engage with YPDC. It covers profile management, membership application, event discovery and registration, volunteering, attendance tracking, certificates, achievements, notifications and feedback — giving every student a single, transparent record of their YPDC journey.

Module specifications

Module	Purpose	Key features
Profile	Maintain personal, academic & contact information with photo and guardian details.	Editable profile • Photo upload • Academic info • Verification status
Membership	Apply for YPDC membership and track the application lifecycle.	Online application • Status tracking • Membership ID & card • Renewal history
Events	Discover all published events with schedules, venues and eligibility.	Event catalogue • Search & filters • Event details • Calendar view
Event Registration	Register for events, receive e-tickets and manage bookings.	One-click registration • QR e-ticket • Waitlist • Cancellation & history
Volunteer Opportunities	Browse and apply for volunteer roles linked to events & projects.	Opportunity board • Applications • Hour logging • Supervisor confirmation
Attendance	View verified attendance across events and volunteer activities.	Event-wise log • QR/biometric/manual marking • Attendance % • Download record
Certificates	Access and download verifiable e-certificates for participation & volunteering.	Certificate wallet • QR verification code • PDF download • Share link
Achievements	Showcase badges, awards and leaderboard points earned through activities.	Badges & trophies • Points system • Leaderboard • Public achievement wall (opt-in)
Notifications	Receive timely alerts for events, approvals, certificates and announcements.	In-app + email/SMS • Preferences • Read/unread centre • Reminders
Feedback	Submit event feedback, ratings and suggestions to improve quality.	Post-event forms • Star ratings • Suggestions box • Response visibility

Key workflows

- ▶ **Membership journey:** Student applies → Admin verifies → ID issued → renewals tracked.
- ▶ **Event journey:** Browse → register → QR ticket → attend (QR scan) → feedback → certificate auto-issued.
- ▶ **Volunteer journey:** Apply → shortlisted → hours logged → supervisor approves → certificate + points.

Portal success metrics

- ▶ **Active students** — monthly KPI on the Admin dashboard and digest reports.
- ▶ **Event registrations** — monthly KPI on the Admin dashboard and digest reports.
- ▶ **Volunteer hours** — monthly KPI on the Admin dashboard and digest reports.
- ▶ **Avg. feedback score** — monthly KPI on the Admin dashboard and digest reports.

5. Portal 2 — Member Portal

For registered YPDC members — duties, tasks & performance • **10 modules**

The Member Portal serves officially inducted YPDC members working under directorates. Beyond student features, it adds a digital Member ID, directorate affiliation, defined responsibilities, task management, duty rosters, performance scoring and activity reporting — making every member accountable, guided and recognised.

Module specifications

Module	Purpose	Key features
Profile & Member ID	Official member profile with a scannable digital ID card.	Digital ID card with QR • Tenure & status • Renewal alerts • ID reprint request
Directorate	Shows the member's assigned directorate, hierarchy and teammates.	Directorate info • Reporting line • Team directory • Transfer requests
Responsibilities	Clearly assigned role description, job duties and tenure goals.	Role card & JD • Tenure objectives • Acknowledgement sign-off
Tasks	Personal to-do list with deadlines, priorities and submissions.	Task inbox • Deadline reminders • File attachments • Status updates
Events / Duties	Event assignments and duty roster with check-in records.	Duty calendar • Role in event • Check-in/out • Duty completion log
Attendance	Meeting and event attendance register with defaulter alerts.	Attendance % • Meeting log • Leave requests • Warning flags
Certificates	Service, appreciation and training certificates issued to the member.	Certificate wallet • Verification QR • Nomination-based awards
Performance	Monthly/quarterly performance score with lead ratings and self-review.	Score dashboard • Lead ratings • Self-assessment • Improvement notes
Reports	Submit periodic activity reports and view submission history.	Report templates • Draft/submit • Review comments • Archive
Notifications	Task, duty, meeting and performance alerts in one centre.	Push/email/SMS • Priority flags • Action links (approve/submit)

Key workflows

- ▶ **Task lifecycle:** Lead assigns → member accepts → submits evidence → lead reviews → scored.
- ▶ **Duty lifecycle:** Roster published → member checks in (QR) → duty logged → hours counted.
- ▶ **Performance cycle:** Monthly auto-score (tasks + attendance + duties) → lead review → published.

Portal success metrics

- ▶ **Active members** — monthly KPI on the Admin dashboard and digest reports.
- ▶ **Task completion %** — monthly KPI on the Admin dashboard and digest reports.
- ▶ **Duty fulfilment %** — monthly KPI on the Admin dashboard and digest reports.
- ▶ **Avg. performance score** — monthly KPI on the Admin dashboard and digest reports.

6. Portal 3 — Directorate Portal

For directorate leadership & teams — planning, execution & accountability • **13 modules**

Each YPDC directorate (e.g. Events, Media, Volunteer Management) operates as a semi-autonomous unit. The Directorate Portal gives directors, assistant directors and team members shared tools for identity, objectives, work planning, tasking, events, proposals, attendance, reporting, documents and performance — so leadership always knows what is planned, what is done, and by whom.

Module specifications

Module	Purpose	Key features
Directorate Profile	Public-facing identity: mandate, scope, contact and branding.	About & mandate • Logo/banner • Contact • Public page link
Director / Assistant Director	Leadership profiles, tenure, authority and contact.	Leadership cards • Tenure dates • Delegation of authority
Team Members	Complete roster with roles, status and joining dates.	Member directory • Role assignment • Active/inactive status
Objectives	Quarterly and annual objectives in OKR style with progress tracking.	Objective setting • Key results • Progress % • Review cycle
Work Plan	Time-bound activity calendar with milestones and owners.	Monthly/quarterly planner • Milestones • Owner mapping • Gantt-style view
Tasks	Create, assign and track tasks across the whole directorate team.	Kanban board • Priorities & deadlines • Comments • Overdue escalation
Events & Activities	Directorate-owned events calendar with ownership and outcomes.	Event pipeline • Ownership • Post-event summary • Media links
Event Proposals	Structured proposal submission routed to Admin for approval.	Proposal form (budget, venue, plan) • Approval workflow • Revision requests
Attendance	Meeting and internal-activity attendance register.	Meeting register • QR/manual marking • Absence reports
Activity Reports	Periodic (weekly/monthly) reports compiled and archived.	Templates • Auto-aggregation of tasks/events • PDF export • Archive
Achievements	Directorate-level awards, milestones and success stories.	Achievement log • Evidence attachments • Public showcase
Documents	Central repository: minutes, letters, plans, policies and MOUs.	Folders & versioning • Access control • Search • Download log
Performance	Directorate scoreboard combining tasks, events, attendance and reports.	Composite score • Trend charts • Inter-directorate ranking • Review notes

Key workflows

- **Proposal flow:** Directorate drafts → Admin reviews → approved/revision → event published → executed → report filed.
- **Planning flow:** Objectives set → work plan derived → tasks assigned → weekly review → monthly report.
- **Governance flow:** Meetings logged → minutes uploaded → action items become tasks → tracked to closure.

Portal success metrics

- **Objectives achieved %** — monthly KPI on the Admin dashboard and digest reports.
- **Events delivered** — monthly KPI on the Admin dashboard and digest reports.
- **Proposal approval rate** — monthly KPI on the Admin dashboard and digest reports.
- **Directorate score** — monthly KPI on the Admin dashboard and digest reports.

7. Portal 4 — Faculty Portal

For faculty mentors & advisors — guidance, supervision & approvals • **9 modules**

Faculty members act as mentors, advisors, judges and approvers. The Faculty Portal recognises their academic identity while defining their YPDC role, connecting them to events and student activities, and routing approval requests (proposals, leaves, certificates) to them with full context — respecting their time while keeping governance strong.

Module specifications

Module	Purpose	Key features
Faculty Profile	Academic and professional profile with photo and credentials.	Qualifications • Experience • Expertise tags • Contact & availability
Department / Designation	Official department, designation and campus affiliation.	Department linkage • Designation history • Campus/branch
YPDC Role	Defined YPDC responsibility: mentor, advisor, judge, patron or coordinator.	Role assignment letter • Scope & tenure • Linked directorate(s)
Events	Invitations for guest sessions, judging panels, trainings and keynote slots.	Invitations inbox • Accept/decline • Session materials • Honorarium record
Student / Member Activities	Supervision log for mentored students, members and projects.	Mentee list • Mentoring sessions log • Project supervision • Remarks
Approvals	One-click approval queue with documents and history.	Pending queue • Approve/return with remarks • E-signature • Audit trail
Reports	Submit mentor reports and view supervised-activity summaries.	Mentor report forms • Supervision summary • Export PDF
Feedback	Give structured feedback on events, members and programs.	Evaluation forms • Ratings • Confidential remarks channel
Notifications	Invitations, approval requests, reminders and announcements.	Priority inbox • Calendar sync • Digest emails

Key workflows

- ▶ **Approval flow:** Request raised → routed to mapped faculty → approve/return with remarks → audit logged.
- ▶ **Mentoring flow:** Mentees assigned → sessions logged → term report submitted → recognised.
- ▶ **Event flow:** Invitation → acceptance → materials shared → session delivered → feedback captured.

Portal success metrics

- ▶ **Sessions delivered** — monthly KPI on the Admin dashboard and digest reports.
- ▶ **Approvals turnaround** — monthly KPI on the Admin dashboard and digest reports.
- ▶ **Mentees guided** — monthly KPI on the Admin dashboard and digest reports.
- ▶ **Mentor rating** — monthly KPI on the Admin dashboard and digest reports.

8. Portal 5 — Admin Portal

For system administrators — control tower of the entire platform • **15 modules**

The Admin Portal is the command centre. It provides a real-time dashboard, complete user/member/directorate/faculty management, event governance, a unified approvals inbox, attendance and certificate control, analytics, notifications, website content, documents, complaints handling and granular roles & permissions — everything needed to run YPDC digitally, transparently and at scale.

Module specifications

Module	Purpose	Key features
Dashboard	Real-time command view: counts, trends, pending items and alerts.	KPI cards • Charts • Pending-approval widget • Quick actions • Activity feed
User Management	Master control of all accounts across every portal and role.	Create/deactivate • Role assignment • Bulk import • Login & audit logs
Member Management	Full member lifecycle: application to induction, transfer, exit and renewal.	Application queue • ID issuance • Transfers • Renewals • Exit & rejoin

Module	Purpose	Key features
Directorate Management	Create and govern directorates, leadership and structures.	Create/merge units • Assign directors • Org chart • Performance comparison
Faculty Management	Onboard faculty, map YPDC roles and track engagement.	Onboarding • Role mapping • Invitations • Engagement history
Events Management	Govern the full event lifecycle from proposal to certificate.	Approve proposals • Publish/unpublish • Registrations • Attendance • Close-out
Applications & Approvals	Single unified queue for every request type in the system.	Membership • Proposals • Leaves • Certificates • Custom forms • SLA tracking
Attendance	Organisation-wide attendance registers, policies and defaulter lists.	Registers • QR/manual/biometric config • Policies • Defaulter reports
Certificates	Design templates, bulk-issue and verify all certificates.	Template designer • Bulk issue • QR verification portal • Revocation
Reports & Analytics	Pre-built and custom reports with exports and scheduled digests.	100+ KPIs • Custom builder • PDF/Excel export • Scheduled emails • Charts
Notifications	Broadcast and targeted multi-channel communication engine.	Templates • Segments • Push/email/SMS • Delivery reports • Announcements
Website Content	Manage all public-portal content without developer help (CMS).	Pages • News • Gallery • Publications • Banners • Menus • SEO fields
Documents	Central, access-controlled document library for the organisation.	Folders • Versioning • Access rights • Retention • Audit of downloads
Feedback / Complaints	Grievance redressal with ticketing, SLA and escalation.	Ticket queue • Categories • SLA timers • Escalation • Resolution & rating
Roles & Permissions	Granular, least-privilege access control for every module & action.	Role builder • 200+ permissions • Delegation • Temporary access • Audit

Key workflows

- ▶ **Approval triage:** All requests land in one queue → auto-routed by type → SLA-monitored → decided → notified.
- ▶ **Event governance:** Proposal → budget/venue check → approve → publish → monitor live → close with report & certificates.
- ▶ **Grievance flow:** Complaint ticketed → acknowledged in 24h → assigned → resolved → complainant rates resolution.

Portal success metrics

- ▶ **Pending approvals** — monthly KPI on the Admin dashboard and digest reports.
- ▶ **SLA compliance %** — monthly KPI on the Admin dashboard and digest reports.
- ▶ **Grievance resolution time** — monthly KPI on the Admin dashboard and digest reports.
- ▶ **Platform adoption %** — monthly KPI on the Admin dashboard and digest reports.

9. Portal 6 — Alumni Portal

For graduates & former members — lifelong network & contribution • **10 modules**

YPDC relationships should not end at graduation or tenure completion. The Alumni Portal keeps alumni connected through rich profiles, a searchable network, mentorship programs, events, achievements, career opportunities and structured ways to volunteer or support — turning former members into lifelong ambassadors, mentors and partners.

Module specifications

Module	Purpose	Key features
Alumni Profile	Lifelong identity carrying forward the member/student history.	Auto-carried history • Photo & bio • Privacy controls • Verified badge
Academic Information	Degrees, institutions, years and specialisations.	Education timeline • Transcripts (optional) • Honours
Professional Information	Current organisation, designation, industry and experience.	Employment timeline • Industry tags • Open-to-work flag
Skills	Searchable skill inventory powering mentoring and hiring matches.	Skill tags • Endorsements • Proficiency levels
Alumni Network	Searchable directory with connection requests and groups.	Directory & filters • Connect/follow • Batch & chapter groups • Messaging
Mentorship	Structured mentor matching between alumni and current students/members.	Mentor registration • Matching • Session scheduling • Feedback & hours
Events	Reunions, networking meetups, talks and homecoming events.	Alumni event calendar • RSVP • Reunion groups • Photo sharing
Achievements	Career and community milestones celebrated organisation-wide.	Achievement posts • Verification • Featured alumni wall
Career Opportunities	Jobs, internships and referrals posted by alumni and partners.	Job board • Referrals • Applications tracking • Employer profiles
Volunteer / Support Opportunities	Give-back channels: guest lectures, funding, CSR and volunteering.	Support catalogue • Pledges • Fund tracking • Impact reports & receipts

Key workflows

- ▶ **Mentorship flow:** Alumni registers as mentor → matching → sessions scheduled → hours & feedback logged.
- ▶ **Career flow:** Opportunity posted → screened → published → applications → shortlist → closure update.
- ▶ **Give-back flow:** Pledge made → acknowledged → utilised with evidence → impact report shared.

Portal success metrics

- ▶ **Registered alumni** — monthly KPI on the Admin dashboard and digest reports.
- ▶ **Mentorship hours** — monthly KPI on the Admin dashboard and digest reports.
- ▶ **Jobs posted/filled** — monthly KPI on the Admin dashboard and digest reports.
- ▶ **Support contributions** — monthly KPI on the Admin dashboard and digest reports.

10. Portal 7 — Guest / Public Portal

For the world — YPDC's public face & front door • **14 modules**

The Guest/Public Portal is YPDC's website — open to everyone without login. It communicates identity (about, vision, leadership, directorates), showcases work (projects, events, news, achievements, gallery, publications) and converts visitors into participants through membership and volunteer registration, contact channels and FAQs. Everything here is managed via the Admin CMS with zero coding.

Module specifications

Module	Purpose	Key features
About YPDC	Founding story, mandate, legal status and organisational overview.	Story page • Facts & figures • Milestones timeline • Partners strip
Vision & Mission	Foundational statements plus values and strategic pillars.	Vision/mission/values • Strategic pillars • SDG alignment

Module	Purpose	Key features
Leadership	Patrons, executives, directors and advisors with profiles.	Leadership grid • Bios • Messages • Tenure info
Directorates	Public directory of all directorates with mandates and highlights.	Directory cards • Mandates • Recent highlights • Contact links
Projects	Flagship and ongoing projects with objectives and progress.	Project pages • Objectives • Timeline • Outcomes • Galleries
Events	Public events calendar: upcoming, ongoing and archived events.	Calendar & list • Details & posters • Registration CTA • Archive
News & Updates	Press releases, announcements and newsletters.	News listing • Article pages • Newsletter signup • Social sharing
Achievements	Organisation, team and individual honours with evidence.	Honour wall • Categories • Evidence & media • Yearly roundups
Gallery	Photos and videos organised by event, project and year.	Albums • Lightbox • Video embeds • Download (optional)
Publications	Reports, magazines, research and policy documents library.	Document library • Categories • Preview & download • Citations
Membership	Public membership information: benefits, criteria, fees and process.	Benefits • Eligibility • Fee table • How-to-apply • Apply CTA
Volunteer Registration	Open volunteer intake form with opportunity linkage.	Registration form • Interest areas • Availability • Auto-acknowledgement
Contact Us	All contact channels plus location map and enquiry form.	Form with ticket ID • Map • Directory • Office hours • Social links
FAQs	Searchable answers reducing support load and confusion.	Categorised FAQs • Search • Ask-a-question • Helpful-vote tracking

Key workflows

- **Conversion flow:** Visitor reads → CTA clicked → membership/volunteer form → ticketed → admin processes → welcome journey.
- **Publishing flow:** Admin drafts content in CMS → preview → publish/schedule → appears instantly on public site.
- **Enquiry flow:** Contact form → auto-ticket + acknowledgement → assigned → responded → feedback on response.

Portal success metrics

- **Monthly visitors** — monthly KPI on the Admin dashboard and digest reports.
- **Membership applications** — monthly KPI on the Admin dashboard and digest reports.
- **Volunteer signups** — monthly KPI on the Admin dashboard and digest reports.
- **Content freshness** — monthly KPI on the Admin dashboard and digest reports.

11. Cross-Cutting Systems (used by all portals)

Authentication, roles & permissions

Access follows **least privilege**: every action maps to a permission; permissions bundle into roles; roles attach to users (a user may hold several, e.g. Member + Volunteer Lead). Sensitive actions (approvals, certificate issuance, data export) require explicit grants and leave an **immutable audit log**.

Role	Sees	Can do (examples)
Guest	Public portal only	Browse, register interest, verify certificates, contact

Role	Sees	Can do (examples)
Student	Student portal	Apply, register, volunteer, feedback, download own certificates
Member	Member + Student	Duties, tasks, reports, duty check-in
Director / AD	Directorate + Member	Plan, assign, propose events, review reports
Faculty	Faculty portal	Mentor, approve, evaluate, report
Alumni	Alumni portal	Network, mentor, post jobs, pledge support
Admin	Everything	Configure, approve, publish, issue, analyse, delegate

Events lifecycle (unified across portals)

- 1. Propose:** Directorate submits proposal (objectives, budget, venue, team) → Admin review → approve / return with remarks.
- 2. Publish:** approved event published to Student/Member/Alumni/Public audiences with eligibility rules and seat caps.
- 3. Register:** one-click registration → QR e-ticket → waitlist when full → reminders before the event.
- 4. Execute:** QR/biometric/manual check-in → live attendance → duty roster enforcement → media capture.
- 5. Close:** attendance frozen → feedback forms opened → certificates auto-issued → activity report filed → analytics updated.

Attendance system

- ▶ **Three capture modes:** QR scan (fast, offline-tolerant), manual register (fallback), biometric/API (where hardware exists).
- ▶ **Policy engine:** per-event and per-directorate rules — late thresholds, minimum % for certificates, defaulter flags.
- ▶ **Transparency:** every user sees their own log; leads see team registers; admin sees organisation-wide heatmaps.

Certificates & achievements

- ▶ **Template designer:** admin creates branded templates with dynamic fields (name, event, date, grade, QR).
- ▶ **Issuance:** automatic (on attendance + feedback completion) or nominated (awards, service) with approval chain.
- ▶ **Verification:** any certificate verified publicly via QR/code — no login needed — with revocation support.
- ▶ **Gamification:** points + badges feed leaderboards at student, member and directorate levels.

Notifications engine

- ▶ **Multi-channel:** in-app + push + email + SMS with per-user preferences and quiet hours.
- ▶ **Templated & translated:** every message type has Urdu + English templates with variable injection.
- ▶ **Actionable:** approval requests, task assignments and tickets carry deep links and one-click actions.
- ▶ **Accountable:** delivery reports per campaign; critical alerts (approvals, SLA breaches) escalate automatically.

Feedback & grievance (complaints)

- ▶ **Event feedback:** structured forms (ratings + open text) open automatically after each event; results feed event scores.
- ▶ **Suggestions:** always-open box per portal; votable; admin triages monthly with public status updates.
- ▶ **Complaints:** ticketed grievance redressal — categories, SLA timers (acknowledge 24h), assignment, escalation, resolution and complainant rating.

Documents & content governance

- ▶ **Single library** with folders, versioning, retention rules and access control; every download logged.
- ▶ **CMS:** all public-portal content editable by authorised admins with draft → preview → publish/schedule and full revision history.
- ▶ **Official outputs:** reports, ID cards, certificates and letters all print/PDF-ready on YPDC letterhead.

12. Data Model — Core Entities (summary)

The platform centres on **people, structure, work and proof**. The table below lists the primary entities; a full ERD with relationships is produced in the design phase. Every entity carries created/updated timestamps, ownership and an audit trail.

Domain	Core entities
People	User, StudentProfile, MemberProfile (ID, tenure, status), FacultyProfile, AlumniProfile, Guardian
Structure	Directorate, Role, Permission, TeamMembership, LeadershipTenure
Engagement	Event, Registration, Ticket, DutyRoster, VolunteerOpportunity, VolunteerApplication, HourLog
Work	Objective (OKR), WorkPlan, Milestone, Task (+comments, attachments), Meeting, Minute
Proof	AttendanceRecord, Certificate (+template, verification code), Badge, PointLedger, Achievement
Governance	Application/ApprovalRequest, Report, Document (+versions), FeedbackResponse, ComplaintTicket (+SLA)
Outreach	NewsPost, Project, Publication, GalleryAlbum, PageContent, FAQ, Enquiry, NewsletterSubscriber
Career/ Alumni	EmploymentRecord, Skill, Connection, Mentorship (+sessions), JobPost, JobApplication, Pledge

Data integrity rules (enforced in v1.0)

One person = one User record (CNIC/email unique). Member IDs are sequential and never reused. Certificates are immutable once issued (revocation creates a new state, never edits). Attendance freezes after event close; corrections need approval with reason. All deletes are soft-deletes.

13. End-to-End Workflows (approval chains)

W1 — Membership application

1. Guest/Student submits application (public or student portal) with documents.
2. Admin verifies eligibility & documents; may request corrections.
3. Approved → Member profile + digital ID issued → directorate allotted → welcome pack notified.
4. Rejected → reason shared; re-application allowed after defined period.

W2 — Event proposal to certificate

1. Directorate submits proposal → Admin (+ mapped Faculty where needed) approves/returns.
2. Event published → registrations → ticketed → reminders.
3. Check-in (QR) → attendance captured → event closed.
4. Feedback collected → certificates auto-issued → report filed → analytics updated.

W3 — Task & duty management

1. Lead creates task/duty with owner, deadline, evidence requirement.
2. Owner accepts → executes → submits evidence → lead reviews → scored/closed.
3. Overdue items escalate automatically; scores feed monthly performance.

W4 — Approvals (generic)

1. Any request enters the unified queue with type, requester, documents and SLA clock.
2. Auto-routed to the mapped approver (Admin/Faculty/Director) → approve / return with remarks.
3. Requester notified at every state change; full history retained for audit.

W5 — Complaint redressal

1. Complaint filed (any portal + public contact) → ticket ID + acknowledgement within 24 hours.
2. Categorised and assigned → investigated → resolved → complainant rates the resolution.
3. Breached SLAs auto-escalate; monthly grievance analytics reviewed by leadership.

14. Reports & Analytics Framework

Every portal contributes data to a shared analytics layer. Admin gets the full catalogue; directors, faculty and members get role-filtered views of their own scope. All reports export to **PDF/Excel** and can be **scheduled by email** (daily/weekly/monthly).

Category	Example reports / KPIs	Audience
Participation	Registrations, footfall, volunteer hours, repeat-participation rate	All leads, Admin
Membership	Applications funnel, approvals, renewals, drop-offs, ID issuance	Admin, Directors
Directorates	Objective completion, work-plan adherence, event delivery, directorate ranking	Leadership
Tasks & duties	Completion %, overdue, escalation count, evidence compliance	Directors, Admin
Attendance	Event-wise %, member-wise %, defaulter lists, trend heatmaps	Leads, Admin
Certificates	Issued/verified/revoked, pending nominations, verification hits	Admin, Public verify
Performance	Member scores, faculty engagement, mentor hours, improvement deltas	Leads, Faculty
Grievance	Tickets by category/SLA, resolution time, satisfaction rating	Leadership
Outreach	Visitors, conversions, news reach, gallery/publication engagement	Media, Admin
Alumni & career	Registrations, mentorship hours, jobs posted/filled, pledges & impact	Leadership

Decision cadence (recommended)

Weekly: pending approvals, overdue tasks, upcoming events. Monthly: directorate scorecards, attendance & grievance review. Quarterly: OKR review, alumni & outreach analysis. Annually: impact report compiled for patrons, partners and the public portal.

15. Non-Functional Requirements

Quality	Requirement (v1.0 target)
Performance	Public pages < 2s; portal pages < 3s at 500 concurrent users; paginated lists everywhere
Availability	99.5% monthly; automated health checks; graceful degradation for QR/offline check-in
Security	Encrypted transport + hashed credentials; role checks server-side; audit logs; session controls; data backups daily
Privacy	Role-scoped data visibility; consent for public showcase; alumni/student privacy controls; retention policy
Usability	Mobile-first responsive; Urdu + English; accessible (WCAG 2.1 AA); max 3 clicks to core actions
Reliability	No silent failures — every submission returns a reference ID; retry-safe forms; idempotent issuance
Scalability	Modular services; Bulk operations (import, issue, notify) queued in background jobs
Maintainability	CMS-managed content; template-managed certificates/notifications; documented APIs for Phase-5 apps
Auditability	Immutable logs for approvals, issuance, permission changes and exports; monthly audit extract

16. Implementation Roadmap (phased delivery)

Delivery is phased so value reaches users early while foundations stay solid. Each phase ends with **user acceptance testing (UAT)** and training for the affected roles.

Phase	Duration*	Delivers	Users unblocked
Phase 1 — Foundation & Public Face	Weeks 1-6	SSO + roles, Admin core (dashboard, users, roles), Guest/Public portal + CMS, contact/FAQ, certificate verification	Public, Admin, Media team
Phase 2 — Engagement Core	Weeks 7-12	Student portal, Events lifecycle (propose→publish→register→QR check-in), Attendance, Notifications, Feedback	Students, Event teams
Phase 3 — Governance & Work	Weeks 13-18	Member + Directorate portals (tasks, duties, work plans, proposals, meetings), Faculty approvals, Certificates & badges, Grievance	Members, Directors, Faculty
Phase 4 — Network & Intelligence	Weeks 19-24	Alumni portal (network, mentorship, jobs), full analytics catalogue, scheduled digests, documents hub, performance scorecards	Alumni, Leadership
Phase 5 — Extensions (optional)	Post-launch	Native mobile apps, LMS/payments integration, advanced BI, chatbot for FAQs	All

*Durations are indicative for a focused team and assume content (logos, copy, photos) is supplied on time. Each phase includes data migration of existing member/event records where available.

Priority order if scope must be trimmed

1. Never cut: identity/roles, event registration + attendance, approvals, certificate verification.
2. Defer first: gamification leaderboards, advanced custom-report builder, SMS (keep email + in-app).
3. Protect: grievance SLA tracking and audit logs — these carry institutional trust.

17. Effort View & Governance

Module count by portal

Portal	Modules	Share
Student Portal	10	12%
Member Portal	10	12%
Directorate Portal	13	16%
Faculty Portal	9	11%
Admin Portal	15	19%
Alumni Portal	10	12%
Guest / Public Portal	14	17%
TOTAL	81	100%

Suggested governance during build

- ▶ **Product owner (YPDC side):** one empowered decision-maker for scope, content and approvals.
- ▶ **Weekly demo:** working software shown every week; feedback logged as tickets, not messages.
- ▶ **Content owners:** one named owner per directorate/public-section for CMS content and media.
- ▶ **Data protection:** staging uses masked data; production access is role-based and logged.

Risks & mitigations

Risk	Mitigation
Scope creep from 81 modules	This report = frozen scope for v1.0; changes via formal change log
Content delays (photos, copy)	CMS training early; placeholder content allowed; owners named upfront
Low adoption / training gaps	Role-based walkthroughs + Urdu video guides + helpdesk in first 60 days
Event-day load spikes	Queued bulk jobs; paginated scans; offline-tolerant QR check-in
Data quality of legacy records	Bulk-import templates with validation; verification step before go-live

18. Conclusion & Recommendations

The YPDC Integrated Portal System, as specified in this report, gives the organisation a **single source of truth** for people, work and proof — from a guest's first visit to an alumnus's lifelong contribution. The design balances **openness** (public showcase, verifiable certificates) with **discipline** (approvals, attendance, audit trails), and balances **ambition** (81 modules) with **pragmatism** (phased delivery, trim priorities).

Recommended next steps

1. **Approve this report (v1.0)** as the frozen functional scope and authorise Phase-1 kickoff.
2. **Nominate owners:** product owner, one content owner per directorate, and the approvals matrix (who approves what).
3. **Supply brand kit & content:** logo, colours, leadership bios/photos, directorate mandates, existing member/event lists.

4. **Confirm policies:** membership criteria & fees, attendance thresholds, certificate rules, grievance SLAs, bilingual requirements.
5. **Kick off Phase 1** with weekly demos; run UAT with real students/members before each phase goes live.

Sign-off block

Prepared for: YPDC Leadership • Document: YPDC Integrated Portal System Report v1.0 •

Date: 10 September 2026

Approved by: _____ Date: _____

Product Owner: _____ Date: _____

19. Appendix A — Master Module Checklist (81 modules)

Use this table to track build & UAT status. Suggested status values: **Not started** / **In progress** / **In UAT** / **Live**.

ID	Portal	Module	Live?
P1	Student	Profile	☐
P1	Student	Membership	☐
P1	Student	Events	☐
P1	Student	Event Registration	☐
P1	Student	Volunteer Opportunities	☐
P1	Student	Attendance	☐
P1	Student	Certificates	☐
P1	Student	Achievements	☐
P1	Student	Notifications	☐
P1	Student	Feedback	☐
P2	Member	Profile & Member ID	☐
P2	Member	Directorate	☐
P2	Member	Responsibilities	☐
P2	Member	Tasks	☐
P2	Member	Events / Duties	☐
P2	Member	Attendance	☐
P2	Member	Certificates	☐
P2	Member	Performance	☐
P2	Member	Reports	☐
P2	Member	Notifications	☐
P3	Directorate	Directorate Profile	☐
P3	Directorate	Director / Assistant Director	☐
P3	Directorate	Team Members	☐
P3	Directorate	Objectives	☐
P3	Directorate	Work Plan	☐
P3	Directorate	Tasks	☐
P3	Directorate	Events & Activities	☐
P3	Directorate	Event Proposals	☐
P3	Directorate	Attendance	☐
P3	Directorate	Activity Reports	☐
P3	Directorate	Achievements	☐
P3	Directorate	Documents	☐
P3	Directorate	Performance	☐
P4	Faculty	Faculty Profile	☐
P4	Faculty	Department / Designation	☐
P4	Faculty	YPDC Role	☐

ID	Portal	Module	Live?
P4	Faculty	Events	☐
P4	Faculty	Student / Member Activities	☐
P4	Faculty	Approvals	☐
P4	Faculty	Reports	☐
P4	Faculty	Feedback	☐
P4	Faculty	Notifications	☐
P5	Admin	Dashboard	☐
P5	Admin	User Management	☐
P5	Admin	Member Management	☐
P5	Admin	Directorate Management	☐
P5	Admin	Faculty Management	☐
P5	Admin	Events Management	☐
P5	Admin	Applications & Approvals	☐
P5	Admin	Attendance	☐
P5	Admin	Certificates	☐
P5	Admin	Reports & Analytics	☐
P5	Admin	Notifications	☐
P5	Admin	Website Content	☐
P5	Admin	Documents	☐
P5	Admin	Feedback / Complaints	☐
P5	Admin	Roles & Permissions	☐
P6	Alumni	Alumni Profile	☐
P6	Alumni	Academic Information	☐
P6	Alumni	Professional Information	☐
P6	Alumni	Skills	☐
P6	Alumni	Alumni Network	☐
P6	Alumni	Mentorship	☐
P6	Alumni	Events	☐
P6	Alumni	Achievements	☐
P6	Alumni	Career Opportunities	☐
P6	Alumni	Volunteer / Support Opportunities	☐
P7	Guest / Public	About YPDC	☐
P7	Guest / Public	Vision & Mission	☐
P7	Guest / Public	Leadership	☐
P7	Guest / Public	Directorates	☐
P7	Guest / Public	Projects	☐
P7	Guest / Public	Events	☐
P7	Guest / Public	News & Updates	☐
P7	Guest / Public	Achievements	☐
P7	Guest / Public	Gallery	☐
P7	Guest / Public	Publications	☐
P7	Guest / Public	Membership	☐
P7	Guest / Public	Volunteer Registration	☐
P7	Guest / Public	Contact Us	☐
P7	Guest / Public	FAQs	☐

20. Appendix B — Glossary

Term	Meaning in this report
YPDC	Youth Peace & Development Council — the organisation this platform serves
Directorate	A specialised unit (e.g. Events, Media) with a director, team, objectives and work plan
Member vs Student	Student = enrolled participant; Member = inducted office-holder under a directorate
Duty	An assigned operational role in a specific event (usher, media, protocol, etc.)
SLA	Service-level agreement — e.g. acknowledge complaints within 24 hours
OKR	Objectives & Key Results — quarterly goal-setting method for directorates
CMS	Content Management System — no-code editing of the public website
QR check-in	Attendance marking by scanning ticket/ID QR codes at venues
UAT	User Acceptance Testing — formal sign-off by real users before go-live